AdifySkills

n8n Automation Mastery
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CLASS NOTES

Module 6 · Production-Grade Engineering

DAY 18 of 25

Error Handling, Sub-Workflows & Reliability

"Production mein workflow ka fail hona = client ka gussa."

⚡ AAJ HUM YEH BANAYENGE

Global Error Workflow → Telegram + Log

Ek baar banao, sab workflows par lagao: koi bhi workflow fail ho to 2 second mein Telegram par alert (workflow, node, error, execution link) aur error log sheet mein entry.

Error Trigger | Execute Workflow | Stop and Error

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Demo aur production mein farak: production workflow fail hone par bhi graceful rehta hai aur aapko turant batata hai.

🎯 Aaj ke baad aap yeh kar paoge

01 Error Trigger & global error workflows
🌍 Station master ko har problem ki khabar.
⚙️ Error Trigger workflow → Settings mein har workflow ka Error Workflow set.

02 Retry on fail, continue on error, fallback paths
🌍 Ghanti nahi baji? Phir bajao.
⚙️ Node Settings: Retry on Fail (3, 5s), On Error: Continue (output).

03 Stop and Error node — custom failures
🌍 Teacher ne bola "galat hai, ruko".
⚙️ IF amount < 0 → Stop and Error "Invalid amount".

04 Sub-workflows: Execute Workflow & reusable modules
🌍 Lego ka ek block baar-baar use.
⚙️ Execute Workflow (sub-workflow) with inputs.

05 Idempotency & duplicate prevention
🌍 Attendance do baar nahi lagti.
⚙️ Sheet/DB mein order_id check → mila to skip.

06 Logging & alerting (Slack/Telegram)
🌍 CCTV + alarm.
⚙️ Slack/Telegram alert: workflow, node, error, execution URL.

⏱ Class ka plan (2 ghante)

10m	35m	45m	20m	10m

`0:00-0:10` **Recap**
🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

`0:10-0:45` **Concept: Reliability Engineering**
🪄 Cycle ki chain utar gayi to tum ro ke nahi baithte — chadhate ho, nahi chadhi to papa ko bulate ho.

`0:45-1:30` **Live Build**
Central error handler + alert + retry queue
🪄 Aaj ki kahani yaad hai — 'Train ka safety system'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

`1:30-1:50` **Chaos Lab**
Hum workflow todenge, aap bachaoge
🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

`1:50-2:00` **Q&A + Homework**
🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Error workflow

Har workflow ki settings mein ek "Error workflow" set karo.

🪄 **Analogy:** Fire alarm

CONCEPT 02

Retry on fail

Node fail ho to X baar, Y second ke gap par retry.

🪄 **Analogy:** Phone na lage to dobara try

CONCEPT 03

Continue on fail

Error ko output mein bhejo aur alag handle karo.

🪄 **Analogy:** Plan B

CONCEPT 04

Sub-workflows

Common logic ek jagah — 10 workflows reuse.

🪄 **Analogy:** Lego blocks

🔑 Aaj ka main idea

Reliability = workflow fail ho to bhi system sambhal le: retry kare, alternate raasta le, duplicate na kare, aur turant batade. Production mein yeh sabse zaroori skill hai.

- API down → 3 baar retry → phir Telegram alert
- Duplicate order webhook → ek hi baar process
- Common "send WhatsApp" sub-workflow sab jagah use

📖 Ek kahani se samjho: Train ka safety system

1 🚆 **Train chal rahi hai** = Workflow run

2 ⚠️ **Signal fail ho gaya** = Node error

3 🔁 **Driver ne dobara try kiya** = Retry on fail

4 🛤️ **Doosri line par moda** = Fallback path

5 📟 **Station master ko alert** = Error Trigger → Telegram

🪄 Ek line mein: “Cycle ki chain utar gayi to tum ro ke nahi baithte — chadhate ho, nahi chadhi to papa ko bulate ho.”

🗺️ Aaj ka workflow: Global Error Workflow → Telegram + Log

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Har client project mein add-on (error monitoring) — aam taur par **₹3,000 – ₹8,000 add-on**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Error Trigger**
   Kisi bhi workflow ka failure
2. ✏️ **Edit Fields (Set)**
   workflow name, node, error message, execution URL
3. ✈️ **Telegram**
   Alert: "❌ Lead Engine fail — HubSpot 401"
4. 📊 **Google Sheets**
   Error log
5. 🔀 **IF**
   Retryable error (429/5xx)?
6. 🧩 **Execute Workflow (Sub-workflow)**
   Retry queue workflow call

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Telegram bot token (BotFather se)**
- ☐ Connect: **Google Sheets (Google OAuth)**
- ☐ Is workflow ko save karo (Active karne ki zaroorat nahi)
- ☐ Har doosre workflow mein: Settings → Error Workflow → "Day 18 — Global Error Workflow" chuno
- ☐ "Errors" tab headers: workflow, node, error, url, time

✅ Test kaise karein

1. Ek test workflow banao jisme HTTP node galat URL par jaaye, uska Error Workflow set karo
2. Use Active karke fail karwao → Telegram alert aaya?

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)

Client bole "kaam nahi hua" tab pata chalna — **120 min**

120 min

→

⚡ Ab (n8n)

2 second mein alert + link — **1 min**

1 min

⏱ Downtime ghante se minute mein. Client ka bharosa bacha.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo 📋 Select karke copy karo

```
❌ {{ $json.workflow.name }} failed at {{ $json.execution.error.node.name }}
{{ $json.execution.error.message }}
🔗 {{ $json.execution.url }}
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
{
  "execution": {
    "id": "2231",
    "url": "https://n8n.you.com/execution/2231",
    "error": {
      "message": "401 Unauthorized",
      "node": {
        "name": "HubSpot"
      }
    }
  },
  "workflow": {
    "name": "Lead Engine"
  }
}
```

📤 Output 📋 Select karke copy karo

```
{
  "alerted": true,
  "logged": true,
  "retry": false
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Ek baar banao, sab workflows par lagao: koi bhi workflow fail ho to 2 second mein Telegram par alert (workflow, node, error, execution link) aur error log sheet mein entry.

Flow: Error Trigger (Koi workflow fail) → Edit Fields (Details nikaalo) → Telegram (Alert) → Google Sheets (Error log)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 **Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Global Error Workflow → Telegram + Log" (nodes: On Any Error, Error Details, Telegram Alert, Error Log).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
4 nodes · ready to import
`adifyskills.com/workflows/day-18-global-error-alert.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/18`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Telegram bot token (BotFather se)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.
"Sheet not found" / column khaali?	Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye.

⚠️ Yeh galtiyan sabse common hain

- ! Error workflow banaya par workflows ki settings mein set nahi kiya
- ! Har error par retry (401 retry karne se theek nahi hota)
- ! Alerts itne zyada ki koi dekhe hi nahi

⭐ Pro tips — time bachane wali tricks

- ★ Idempotency key (order ID) se duplicate side-effects roko
- ★ Sub-workflow inputs ko clearly define karo
- ★ Weekly error report — trends dekho

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Global error workflow: kisi bhi workflow fail par Telegram alert with error details + execution link.

📝 Homework

Apne purane 3 workflows mein proper error handling + sub-workflows add karo.

☐ Homework ho gaya

BEGINNER
Error workflow → Telegram ☐

INTERMEDIATE
Retry on fail + fallback path ☐

ADVANCED
Dead-letter queue in Postgres + replay ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Same workflow 10 minute mein 3 baar fail ho to alert mein "🔥 Repeated failure" likha aaye.

💡 Hint 1
Pichhle errors Error Log sheet mein hain — unhe padh sakte ho.

💡 Hint 2
Sheets read → Code mein last 10 min ke same workflow errors gino.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Global failure handler ka trigger?

(A) Manual
(B) Form
(C) Error Trigger
(D) Webhook

Q2. Kaunsa error retry karne layak?

(A) 401
(B) 400 validation
(C) 404
(D) 429 / 503

Q3. Reusable logic ke liye?

(A) Sub-workflow
(B) Copy-paste
(C) Sticky note
(D) Pin

Q4. Custom failure throw karne ka node?

(A) Sort
(B) Stop and Error
(C) NoOp
(D) Merge

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Execution
Workflow ka ek baar chalna. Har run ka record Executions tab mein dikhta hai.

Retry
Fail hone par apne aap dobara try karna.

Sub-Workflow
Ek workflow jo doosre workflow ke andar se chalaya jaata hai — reusable block.

Idempotency
Ek hi kaam do baar aaye to bhi result ek hi baar ho (duplicate se bachna).

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 19

Self-Hosting n8n: Docker, VPS & Domains

"Apna khud ka n8n server — unlimited workflows."

☐ **Day 18 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → C
Error Trigger

Q2 → D
429 / 503

Q3 → A
Sub-workflow

Q4 → B
Stop and Error

✅ Challenge ka solution

Error Details → Google Sheets (read Errors) → Code: count rows where workflow same and time > now-10min → Telegram text mein count >= 3 ho to 🔥 prefix.

📚 Official docs

- Error handling ↗
- Sub-workflows ↗
- Execute Workflow ↗
- Error Trigger ↗
- Workflow settings ↗

▶ Videos

- n8n error handling workflow ↗
- n8n sub workflows tutorial ↗

Doubt poochho
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Day 18 lesson page
Quiz, copy buttons, videos
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